

DUCK PIER, NC, USA

WMO: 998209

Lat: 36.184N

Lon: 75.746W

Elev: 7

StdP: 101.24

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.2	(c) -2.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 17.1	(k) 6.0	(l) 15.7	(m) 3.2	(n) 8.4	(o) 330	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.3	(c) 31.4	(d) N/A	(e) 30.2	(f) N/A	(g) 29.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 5.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.3	(b) 14.2	(c) 12.2	(e) -6.6	(f) 33.6	(g) 2.6	(h) 1.2	(i) -8.4	(j) 34.4	(k) -9.9	(l) 35.1	(m) -11.4	(n) 35.7	(o) -13.2	(p) 36.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	6.0	6.9	9.9	14.7	19.5	24.2	26.1	25.8	23.5	18.9	13.3	9.4		
	DBStd	8.04	5.38	4.26	4.32	4.14	3.78	2.85	2.02	1.82	2.18	3.56	3.75	4.54		
	HDD10.0	398	148	104	58	7	0	0	0	0	0	1	13	67		
	HDD18.3	1592	383	320	261	123	33	1	0	0	0	36	156	279		
	CDD10.0	2794	23	18	56	148	293	427	498	489	406	278	111	47		
	CDD18.3	947	0	1	1	14	68	178	240	231	156	54	4	1		
	CDH23.3	6168	0	0	1	35	254	1169	2086	1839	680	103	2	0		
	CDH26.7	1172	0	0	0	1	37	261	476	327	66	5	0	0		
(14) Wind	WSAvg	6.2	6.6	6.5	7.1	6.6	6.5	5.8	5.2	5.1	6.2	6.3	6.5	6.2		
(15) Precipitation	PrecAvg	1404	109	93	111	95	110	128	158	153	144	106	90	106		
	PrecMax	1872	191	197	224	178	204	239	254	291	506	222	235	218		
	PrecMin	979	45	36	32	34	42	46	93	60	36	3	21	43		
	PrecStd	187	36	41	38	35	42	47	38	54	98	49	52	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	20.4	22.6	26.0	29.8	32.6	33.4	32.5	30.1	27.1	23.0	21.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	18.2	18.3	20.3	24.4	27.5	30.8	31.6	30.6	28.4	25.5	21.4	19.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	16.5	16.4	18.8	22.9	26.0	29.5	30.2	29.4	27.2	24.5	19.9	18.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	14.0	13.7	17.0	21.1	24.6	28.0	29.1	28.4	26.3	23.5	18.5	16.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	5.9	6.2	6.1	5.3	4.6	4.3	3.6	3.0	4.1	4.9	5.4		
		MCDBR	8.5	9.3	9.4	8.9	7.4	6.6	6.1	5.5	4.8	4.9	6.5	7.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.329	0.353	0.396	0.440	0.486	0.514	0.502	0.416	0.366	0.333	0.325		
	taud		2.537	2.508	2.467	2.356	2.271	2.193	2.123	2.145	2.387	2.511	2.546	2.551		
	Ebn at Noon		879	912	916	887	849	807	781	782	839	859	856	850		
	Edh at Noon		79	91	104	122	135	146	155	149	112	90	78	74		
(44) All-Sky Solar Radiation	RadAvg		2.35	3.13	4.22	5.39	6.07	6.43	6.09	5.52	4.56	3.64	2.66	2.12		
	RadStd		0.17	0.31	0.32	0.38	0.45	0.36	0.30	0.36	0.31	0.36	0.19	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+113

Nomenclature: See separate page